C3.3 What are electrolytes and what happens during electrolysis?			
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*		***Linked learning opportunities***
Electrolysis is used to extract reactive metals from their ores. Electrolysis is the decomposition of an electrolyte by an electric current. Electrolytes include molten and dissolved ionic compounds. In both cases the ions are free to move. During electrolysis non-metal ions lose electrons to the anode to become neutral atoms. Metal (or hydrogen) ions gain electrons at the cathode to become neutral atoms. **The addition or removal of electrons can be used to identify which species are reduced and which are oxidised. These changes can be summarised using half equations**. Electrolysis is used to extract reactive metals from their molten compounds. During the electrolysis of aluminium, aluminium oxide is heated to a very high temperature. Positively charged aluminium ions gain electrons from the cathode to form atoms. Oxygen ions lose electrons at the anode and form oxygen molecules which react with carbon electrodes to form carbon dioxide. The process uses a large amount of energy for both the high temperature and the electricity involved in electrolysis. Some extraction methods, such as the recovery of metals from waste heaps, give a dilute aqueous solution of metals ions. When an electric current is passed through an aqueous solution the water is electrolysed as well as the ionic compound. Less reactive metals such as silver or copper form on the negative electrode. If the solution contains ions of more reactive metals, hydrogen gas forms from the hydrogen ions from the water. Similarly, oxygen usually forms at the positive electrode from hydroxide ions from the water. A concentrated solution of chloride ions forms chlorine at the positive electrode.	1.	describe electrolysis in terms of the ions present and reactions at the electrodes	**Practical work:** • investigate what type of substances are electrolytes
	2.	predict the products of electrolysis of binary ionic compounds in the molten state	
	3.	recall that metals (or hydrogen) are formed at the cathode and non-metals are formed at the anode in electrolysis using inert electrodes	
	4.	**use the names and symbols of common elements and compounds and the principle of conservation of mass to write half equations**	
	5.	**explain reduction and oxidation in terms of gain or loss of electrons, identifying which species are oxidised and which are reduced**	
	6.	explain how electrolysis is used to extract some metals from their ores including the extraction of aluminium	
	7.	describe competing reactions in the electrolysis of aqueous solutions of ionic compounds in terms of the different species present, including the formation of oxygen, chlorine and the discharge of metals or hydrogen linked to their relative reactivity	**Practical work:** • investigate the effects of concentration of aqueous solution, current, voltage on the electrolysis of sodium chloride
	8.	describe the technique of electrolysis of an aqueous solution of a salt *PAG2*	